

IINTS-AF EUCYS Evidence Pack

Competition evidence bundle for EUCYS review

Project: IINTS-AF SDK

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Competition: EUCYS 2026

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Status: Jury-facing explanation pack built from the maintained SDK repository

This PDF was rendered from the markdown report and packaged benchmark artifacts inside the IINTS-AF SDK workspace.

What This Pack Is

This pack explains IINTS-AF as a scientific software project, not just as a demo. It is designed for judges who need to understand three things quickly:

- what problem the SDK solves
- how the SDK works technically
- why glucose physiology and data quality make this problem hard

The full benchmark report is still the primary results document. This evidence pack is the companion set of short PDFs that explain the system around the numbers.

Recommended Reading Order

Order	Document	Best for
1	EUCYS_00_INDEX.pdf	Fast orientation
2	EUCYS_02_PHYSIOLOGY_AND_DATA_BRIEF.pdf	Why diabetes simulation is difficult
3	EUCYS_01_TECHNICAL_BRIEF.pdf	How the SDK is built
4	EUCYS_03_IMPACT_ETHICS_AND_MAINTENANCE.pdf	Why the project matters and how it is governed
5	EUCYS_04_JURY_QA.pdf	Fast answers during judging
6	EUCYS_05_PHYSIOLOGY_REFERENCE_BROCHURE.pdf	Standalone physiological reference for doctors and engineers
7	EUCYS_06_JURY_PHYSIOLOGY_BRIEF.pdf	Fast handout during a live conversation
8	research/EUCYS_REPORT.pdf	Full benchmark results

One-Minute Project Story

Type 1 diabetes requires constant decision-making around glucose, food, insulin, exercise, sensor data, and uncertainty. A wrong insulin decision can be dangerous, so research software in this domain should be reproducible, auditable, and safety-aware.

IINTS-AF is an open-source SDK for testing insulin decision systems on virtual patients. It provides simulation, baseline comparison, data-quality validation, safety supervision, result packaging, and edge-device workflows. The goal is not to replace clinicians or medical devices. The goal is to make pre-clinical algorithm research more transparent and easier to review.

The core idea is simple:

Before an insulin algorithm is trusted, it should be tested against realistic virtual-patient scenarios, corrupted data, safety ablations, and reproducible baselines.

What Is New In This Project

IINTS-AF combines components that are often separated across research prototypes:

Layer	What IINTS-AF Adds
Simulation	Virtual patients, meals, exercise, scenarios, run bundles
Algorithm comparison	Candidate controller plus explicit baselines
Data quality	MDMP certification and realism checks before training/evaluation
Safety	Deterministic supervisor guarding risky decisions
Reproducibility	Fixed seeds, protocol bundles, manifests, exact CLI paths
Review	Markdown/PDF reports, CSV tables, figures, judge-facing packs
Edge deployment	Raspberry Pi, UNO Q, Jetson-oriented workflows for demonstrators

Final Benchmark Snapshot

The final EUCYS benchmark bundle executed 3600 locked runs across profiles, scenarios, algorithms, study arms, and seeds.

Metric	Result
Total runs	3600
Overall mean Time in Range	81.42%
95% CI for overall TIR	81.09 to 81.75
Clean-arm candidate TIR	93.04%
Clean vs corrupted TIR delta	+17.11 points
Python test suite after pack work	369 passed, 4 skipped
GitHub Actions after latest push	Python CI, Health Badges, Notebook Bake all succeeded

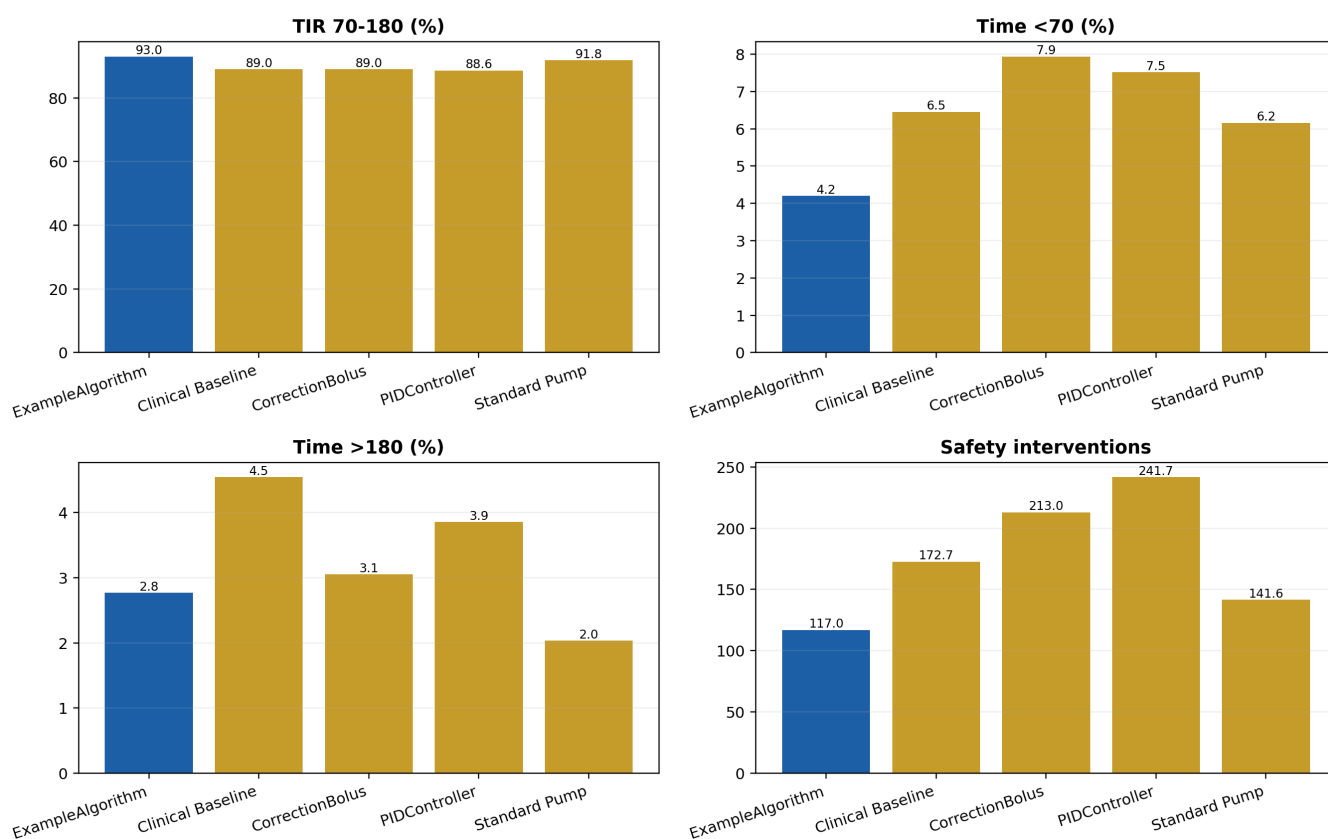
EUCYS Main Figure - Candidate vs Baselines (Clean Certified Arm)

Figure: EUCYS main benchmark figure

How To Verify The Main Evidence

The final benchmark can be reproduced with:

```
tools/research/run_eucys_final.sh \  
  --algo algorithms/example_algorithm.py \  
  --output-dir results/eucys_2026 \  
  --seeds 1,2,3,4,5,6,7,8,9,10 \  
  --no-prepare-ai
```

The report PDF can be regenerated with:

```
tools/research/render_eucys_report_pdf.sh
```

This pack can be regenerated with:

```
tools/research/build_eucys_pack.sh
```

What This Project Does Not Claim

IINTS-AF is not a medical device. It does not prescribe insulin to real patients. It is not a replacement for clinical trials or certified automated insulin delivery systems.

The correct claim is narrower and stronger:

IINTS-AF is a reproducible open research platform for testing and explaining insulin decision algorithms before they ever reach real-world deployment.

Source Backbone

The pack uses the source legend already maintained in docs/EVIDENCE_BASE.md, including ADA Standards of Care 2026, the international Time in Range consensus, OhioT1DM, physiological simulation literature, UVA/Padova simulator literature, and recent work on generative type 1 diabetes simulators.